CS201 Midterm 1 Part A

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I. Question (Algorithm Analysis) (20 pts)

```
int algorithm1(int N){
    if (N == 0)
        return 0;
    int sum = 0;
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            sum++;
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            for (int k = 0; k < 2; k++)
                if ((i + j + k) % 3 == 1)
                    sum += algorithm1(N / 2) + algorithm1(N / 2);
                else
                    if ((i + j + k) % 3 == 0)
                        sum += algorithm1(N / 2);
                    else
                        sum++;
    return sum;
}
```

What is the time complexity of algorithm1? Solve the recurrence equation without the master theorem.

II. Question (Algorithm Analysis) (20 pts)

```
int algorithm2(int N){
    int i, j, k, sum = 0;
    for (i = 1; i <= N - i; i++)
        sum++;
    for (i = 1; i <= N; i++)
        for (j = 1; j <= 2* i; j++)
            for (k = 1; k <= j - i; k++)
                sum++;
    for (i = 0; i <= N; i++)
        for (j = i + 1; j <= N ; j++)
            sum++;
    return sum;
}
```

What does the function algorithm2 return in terms of N? Assume that N is even.